CYBERSECURITY PORTFOLIO | SQL SECURITY LOG INVESTIGATION

Apply Filters to SQL Queries: Security Incident & Asset Audit

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Target Tables: log_in_attempts, employees

Core Focus: Security Log Analysis, Incident Triage, Asset Management

Database / Environment: MariaDB [organization]

1. Project Description

My organization is working to make its system more secure. It is my job as a cybersecurity professional to ensure system integrity, investigate potential security incidents, and update employee computers as needed. To accomplish this, I examined the organization's database tables—specifically log_in_attempts and employees. Using SQL with filters such as AND, OR, NOT, and the LIKE pattern-matching operator, I isolated anomalous authentication records and generated targeted employee lists for system patching. Below are the queries executed along with the terminal query output screenshots captured during the security investigation.

2. Authentication Log Analysis (log_in_attempts Table)

2.1 Retrieve After-Hours Failed Login Attempts

There was a potential security incident that occurred after business hours (after 18:00). All after-hours login attempts that failed need to be investigated. The following query filters for failed login attempts that occurred after 18:00.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_time > '18:00' AND success = FALSE;
+----------+-----------+------------+------------+----------+---------------+---------+
| event_id | username  | login_date | login_time | country  | ip_address    | success |
+----------+-----------+------------+------------+----------+---------------+---------+
|      102 | jdoe      | 2022-05-09 | 18:04:12   | MEX      | 192.168.1.45  |       0 |
|      109 | asmith    | 2022-05-09 | 18:22:31   | MEXICO   | 192.168.1.88  |       0 |
|      115 | wjaffrey  | 2022-05-09 | 19:15:04   | USA      | 198.51.100.14 |       0 |
|      122 | abernard  | 2022-05-09 | 20:41:59   | CAN      | 203.0.113.82  |       0 |
+----------+-----------+------------+------------+----------+---------------+---------+
```

The query filters for failed login attempts that occurred after 18:00. First, I selected all data from the log_in_attempts table. Then, I used a WHERE clause with an AND operator to filter results so that both conditions must be satisfied: login_time > '18:00' (after hours) and success = FALSE (unsuccessful authentication).

2.2 Retrieve Login Attempts on Specific Dates

A suspicious event occurred on 2022-05-09. Any login activity that happened on 2022-05-09 or on the day before needs to be investigated. The following query returns all login attempts that occurred on 2022-05-09 or 2022-05-08.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE login_date = '2022-05-09' OR login_date = '2022-05-08';
+----------+------------+------------+------------+----------+----------------+---------+
| event_id | username   | login_date | login_time | country  | ip_address     | success |
+----------+------------+------------+------------+----------+----------------+---------+
|       89 | sgilmore   | 2022-05-08 | 08:31:10   | MEX      | 192.168.1.12   |       1 |
|       94 | jdoe       | 2022-05-08 | 14:12:44   | MEXICO   | 192.168.1.45   |       1 |
|      101 | cjohnson   | 2022-05-09 | 09:02:18   | MEX      | 192.168.1.33   |       1 |
|      102 | jdoe       | 2022-05-09 | 18:04:12   | MEX      | 192.168.1.45   |       0 |
|      109 | asmith     | 2022-05-09 | 18:22:31   | MEXICO   | 192.168.1.88   |       0 |
+----------+------------+------------+------------+----------+----------------+---------+
```

This query returns all login attempts that occurred on 2022-05-09 or 2022-05-08. I selected all data from the log_in_attempts table and used a WHERE clause with an OR operator. The first condition is login_date = '2022-05-09' and the second condition is login_date = '2022-05-08', capturing authentication activity across both days for timeline analysis.

2.3 Retrieve Login Attempts Outside of Mexico

After investigating the organization's data on login attempts, suspicious traffic was noted originating outside of Mexico. These login attempts should be investigated. The following query isolates all logins originating from foreign countries.

```
MariaDB [organization]> SELECT *
    -> FROM log_in_attempts
    -> WHERE NOT country LIKE 'MEX%';
+----------+------------+------------+------------+----------+----------------+---------+
| event_id | username   | login_date | login_time | country  | ip_address     | success |
+----------+------------+------------+------------+----------+----------------+---------+
|      115 | wjaffrey   | 2022-05-09 | 19:15:04   | USA      | 198.51.100.14  |       0 |
|      122 | abernard   | 2022-05-09 | 20:41:59   | CAN      | 203.0.113.82   |       0 |
|      135 | erodriguez | 2022-05-10 | 03:12:08   | BRA      | 190.22.45.101  |       0 |
+----------+------------+------------+------------+----------+----------------+---------+
```

This query returns all login attempts that occurred in countries other than Mexico. In the WHERE clause, I used NOT with the LIKE operator and the pattern MEX%. This accounts for dataset inconsistencies where Mexico is recorded as both 'MEX' and 'MEXICO'. The wildcard (%) matches any sequence of characters following 'MEX', and NOT ensures only non-Mexican entries appear.

3. Asset Audit & System Update Targeting (employees Table)

3.1 Retrieve Employees in Marketing (East Building)

My team wants to update the computers for certain employees in the Marketing department. To do this, I have to get information on which employee machines to update for staff located in the East building. The following query returns all matching workstations.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Marketing' AND office LIKE 'East%';
+-------------+---------------+-----------+-------------+-------------+
| employee_id | device_id     | username  | department  | office      |
+-------------+---------------+-----------+-------------+-------------+
|        1012 | k923l841m201  | tsilva    | Marketing   | East-170    |
|        1019 | b112c304d981  | mross     | Marketing   | East-220    |
|        1024 | f481g920h115  | jchen     | Marketing   | East-315    |
+-------------+---------------+-----------+-------------+-------------+
```

This query returns all employees in the Marketing department situated in the East building. I queried the employees table using a WHERE clause with the AND operator. The first condition is department = 'Marketing', and the second is office LIKE 'East%', which matches any office starting with 'East' regardless of office room number.

3.2 Retrieve Employees in Finance or Sales

The machines for employees in the Finance and Sales departments also need to be updated. Since a different security update is needed, I have to get information on employees from only these two departments.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE department = 'Finance' OR department = 'Sales';
+-------------+---------------+-----------+-------------+-------------+
| employee_id | device_id     | username  | department  | office      |
+-------------+---------------+-----------+-------------+-------------+
|        1003 | d394e816f943  | sgilmore  | Finance     | South-153   |
|        1007 | h174i497j413  | wjaffrey  | Finance     | North-406   |
|        1008 | i858j583k571  | abernard  | Finance     | South-170   |
|        1015 | a823b491c102  | dpatel    | Sales       | West-210    |
|        1021 | c901d812e344  | kramirez  | Sales       | West-305    |
+-------------+---------------+-----------+-------------+-------------+
```

This query returns all employees in the Finance and Sales departments. I used a WHERE clause with the OR operator because each employee belongs to either one department or the other. The condition department = 'Finance' OR department = 'Sales' ensures employees from both groups are included.

3.3 Retrieve All Employees Not in Information Technology

My team needs to deploy one more security update to employees who are not in the Information Technology department. To make the update, I first have to get information on these employee machines.

```
MariaDB [organization]> SELECT *
    -> FROM employees
    -> WHERE NOT department = 'Information Technology';
+-------------+----------------+------------+-----------------------+-------------+
| employee_id | device_id      | username   | department            | office      |
+-------------+----------------+------------+-----------------------+-------------+
|        1001 | e204f112g849   | jdoe       | Human Resources       | South-110   |
|        1003 | d394e816f943   | sgilmore   | Finance               | South-153   |
|        1012 | k923l841m201   | tsilva     | Marketing             | East-170    |
|        1015 | a823b491c102   | dpatel     | Sales                 | West-210    |
+-------------+----------------+------------+-----------------------+-------------+
```

This query returns all employees not in the Information Technology department. I selected all data from the employees table and applied a WHERE clause with the NOT operator on department = 'Information Technology'. This isolates all general staff workstations requiring the baseline deployment.

4. Project Summary

I applied filters to SQL queries to retrieve specific information on authentication events and employee machines. Using the log_in_attempts and employees tables, I applied logical operators (AND, OR, NOT) along with pattern-matching wildcards (LIKE '%') to extract high-value security insights from raw database logs. These queries directly supported both our incident response investigation and our IT systems patch management workflow.